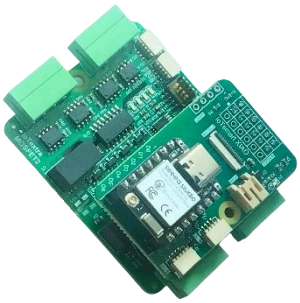
MAKE THINGS HAPPY !

8 MOSFET Outputs (SOP-8) - IoTextra MOSFET2

MOSFET Output Module



8 MOSFET Outputs (SOP-8) - IoTextra MOSFET2 Module (hereinafter **IoTextra MOSFET2**, version 3.02) provides eight independent, galvanically isolated MOSFET switch channels. The module supports two 4-channel isolation groups, each powered by a dedicated isolated DC-DC converter and uses a 4-channel optocoupler.

The maximum switching voltage and current depend on the MOSFET model installed and are listed in the table below:

Vds, V	Vid, A
90	6
60	7
30	9

The module provides LED indicators for the status of each MOSFET channel.

Up-to-date technical information about the module is available on GitHub.

SOFTWARE POWER:

The **IoTextra MOSFET2** module offers native support for popular IoT platforms:

This hardware compatibility unlocks the full potential of the **IoTflow platform**. It enables you to:

- Create custom **Node-RED** nodes effortlessly.
- Integrate via **MQTT** with third-party global platforms like **Blynk**.
- Scale your project from a single node to a complex industrial network.

There are two modes for using the **IoTextra MOSFET2** module:

- ***GPIO Mode***. In this mode, the MOSFET state is controlled using the **AP0-AP7** signals in the **HOST-P** connector.
- ***I²C Mode***. In this mode, the MOSFET state is controlled via the **I²C** bus using the I/O expander installed on the module (TCA9534 or a compatible one). Up to 16 modules can be connected to one **I²C** bus.

Combined Use: It is also possible to use the **AP0-AP7** signals for some channels through the **HOST-P** connector and for the rest via the I/O expander.





This module is designed for systems requiring high-speed switching and precise control of multiple DC loads with significant current demands (up to 9A depending on the model).

APPLICATIONS:

🏭 **Industrial Automation (Drive and Valve Control)**

MOSFET-based outputs provide high-speed PWM control for adjusting DC motor speeds or heating element intensity. The module is ideal for controlling solenoid valves in dosing systems where high switching frequency and zero mechanical wear are essential.

Recommended software: Node-RED, IoTflow

🔔 **IIoT (Status Indication and Alerting)**
8 channels allow for the management of extensive LED panel networks, high-power sounders, and signal towers. Low RDS(on) minimizes heat dissipation, ensuring stable operation within enclosed industrial cabinets even under continuous heavy loads.
Recommended software: Tasmota

🌱 **Smart Farming (Greenhouse and Irrigation Automation)**
The module controls high-power LED grow lights with smooth sunrise/sunset dimming and operates pumps for micro-drip irrigation. Its high voltage capability (up to 90V depending on the model) allows it to work with a wide range of DC equipment and power systems.
Recommended software: Node-RED, IoTflow

🏠 **Smart Home (Ambient Lighting and Access Control)**
Used for multi-channel control of high-power RGBW strips and electromagnetic locks. MOSFET outputs ensure completely silent operation, which is critical for residential environments compared to the audible clicking of mechanical relays.
Recommended software: Home Assistant

FEATURES:

- Compatibility with major known microcontrollers
- Module power supply is 5VDC, which feeds a 3V3 regulator for powering the logic section
- 3V3 Indicator LED
- Protection against reverse power supply polarity
- 8 independent N-channel MOSFET switch channels
- Two 4-channel isolated MOSFET groups
- Channel control via 4-channel optocoupler, for gate drive isolation
- Two isolated DC-DC converters provide group-level power isolation for the MOSFET gate drive circuits
- Vds and Vid depend on the type of MOSFET installed:

Vds, V	Vid, A
90	6
60	7
30	9

- RDS(on) < 20mΩ
- When working with the module: logical “1” (2.5–5V) is the MOSFET off (closed); logical “0” (0–1.1V) is the MOSFET on (open)
- Module features LED indicators for the MOSFET channel status: the LED lights up when the MOSFET is on
- The MOSFET is controlled using **AP0-AP7** signals via the **HOST-P** connector and/or over the **I²C** serial bus via an I/O expander
- The I/O expander used in the module is a **TCA9534** or a compatible chip
- The expander's **I²C** address (**A2-A0**) is set using DIP microswitch:

TCA9534	0	1	0	0	A2	A1	A0	x
TCA9534A	0	1	1	1	A2	A1	A0	x

- Connection to the module via the **I²C** bus is done through **Qwiic©** connectors or through pins 11 (**SDA**) and 12 (**SCL**) of the **HOST-P12** connector
- Transient suppression and electrostatic discharge protection (ESD protection) of signals on Qwiic connectors is done using a TVS diode assembly
- Module size is 47x56 mm. The module has mounting holes allowing it to be installed in the base module or on a Raspberry Pi

DC-DC CONVERTER OPTIONS:

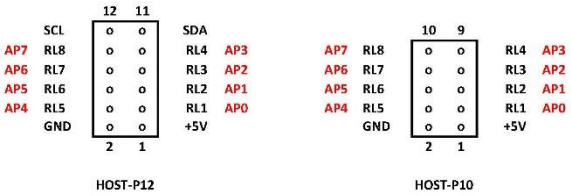
Two isolated DC-DC converters provide group-level power isolation for the MOSFET gate drive circuits. Four converter configurations are supported (one or two units per isolation group):

Configuration	Type	Notes
low-profile (×1)	5V → 5V	Single converter, R55/R56 (0Ω)
low-profile (×2)	5V → 5V	Two converters, one per group
standard (×2)	5V → 5V	Two converters, one per group
standard (×2)	5V → 12V	For 12V gate drive MOSFETs

HOST-P CONNECTOR:

Depending on the version of the Depending on the version of the **IoTextra MOSFET2** module can have either a 12-pin (**HOST-P12**) or a 10-pin (**HOST-P10**) **HOST-P** connector.

The pinouts for these connectors are shown in the image below.



The **HOST-P** connector is used differently depending on how the microcontroller interacts with the **IoTextra MOSFET2** module:

1) ***Standalone Use:*** In this mode, the MOSFET control are available through the **HOST-P10** (**HOST-P12**) connector or via the **I²C** bus using the I/O expander.
2) ***Smart Use:*** In this mode, an **IoTsmart** microcontroller module is inserted into the **HOST-P12** connector. MOSFET control and the 5V module power supply is done through the **HOST-P12** connector. Power is supplied from the **IoTsmart** module. Below is a photo of the **IoTextra MOSFET2** module with the **IoTsmart** module:



IoTextra MOSFET2 with **IoTsmart RP2350A**



IoTextra MOSFET2 with **IoTsmart ESP32-S3**

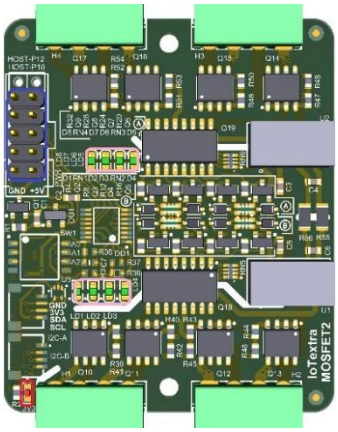
3) ***Mezzanine Use:*** In this mode, the **IoTextra MOSFET2** module is installed in the base module and the MOSFET control and the 5V module power supply are done through the **HOST-P12** connector. Photo of the **IoTextra MOSFET2** module installed in **IoTbase** modules below.



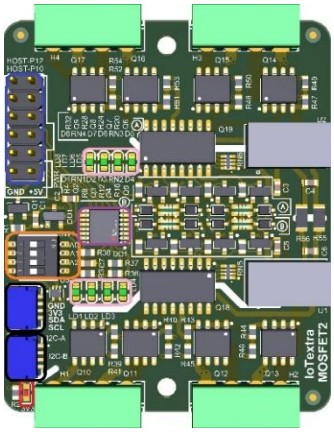
IoTextra MOSFET2 and **IoTbase Feather** with **Particle Proton2** installed

COMPONENT LAYOUT:

There are three versions of the **IoTextra MOSFET2** module, and accordingly, each with a different **top-side** component layout:



HOST-P10

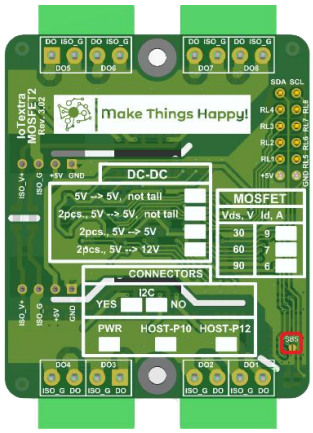


HOST-P12 and Qwiic©

It is possible to control MOSFET state through the **HOST-P** connector (highlighted in blue).
The **Qwiic©** connectors are highlighted in black.
The I/O expander is highlighted in purple.
The DIP microswitch for setting the I/O expander's **I²C** address is highlighted in orange. The “ON” position of the microswitch corresponds to a “0” value in the respective address bit (A0, A1, and A2) for the I/O expander. The “OFF” position corresponds to a “1” value.
The MOSFET status indicators are also on the **top-side** and are highlighted in pink.
There is a 3V3 power indicator. The power indicator are highlighted in red in the **top-side** image.

JUMPERS:

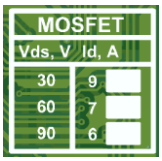
The image below shows the component and marking layout on the **bottom-side** of the **IoTextra MOSFET2** module:



The **bottom-side** of the **IoTextra MOSFET2** module has a ***jumper that allows powering devices connected through the Qwiic© connector***. It is highlighted in red. By default, the jumper is absent.

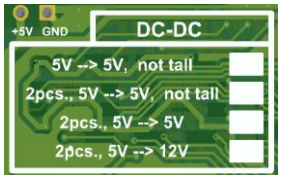
CONFIGURATION TABLES:

The **bottom-side** of the module provides information about the MOSFET2 types used:



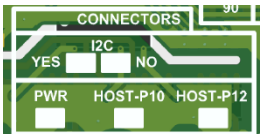
Vds, V	Vid, A
90	6
60	7
30	9

On the **bottom side** there is a configuration table that defines the installed DC-DC converters:



Configuration	Type	Notes
low-profile (×1)	5V → 5V	Single converter, R55/R56 (0Ω)
low-profile (×2)	5V → 5V	Two converters, one per group
standard (×2)	5V → 5V	Two converters, one per group
standard (×2)	5V → 12V	For 12V gate drive MOSFETs

It also provides information on whether the following connectors are installed:



- **Qwiic©** connectors for **I²C**
- **PWR** power connector
- **HOST-P10** or **HOST-P12** connector for relays control

EXTERNAL SIGNALS CONNECTION:

To connect the load to the MOSFET, the **H1 – H4** connectors are used. Removable terminal blocks with a 3.5mm pitch are recommended for these connectors. The pin layout is marked on the module's **bottom-side**.

ACCESSORIES:

The following accessories may be required for using the module:

- A set of four removable terminal blocks with a 3.5mm pitch for terminal blocks **H1-H4**
- A set of two standoffs and four screws for mounting the module into the **IoTbase** series base module.
- Cable for the **HOST-P10** connector
- Cable for the **Qwiic©** connector
- Cable for the **PWR** connector